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EasyDiffraction and crysta for diffraction data analysis at ESS

Content

EasyDiffraction and crysta are being developed to support diffraction data analysis during ESS hot commissioning and early science. The aim is to make existing analysis methods easier to use and reduce the time needed for structure refinement.

EasyDiffraction provides a common Python interface to the crystallographic libraries CrysFML, CrysPy and PDFfit2, which serve as calculation engines. Command-line and notebook workflows are available. Depending on the calculation engine, it supports powder and single crystal diffraction with neutrons and X-rays, pair distribution function analysis, multiple structures and datasets, single, joint and sequential fitting, and Bayesian analysis. Tutorials with data from several facilities are available for training and testing.

crysta is a new C++ calculation and refinement engine with Python bindings. It supports powder diffraction and neutron single crystal refinement, with further single crystal capabilities under development. Development uses AI under human supervision, with unit, integration and system tests and comparisons against independent programs and libraries, including FullProf and CrysPy. In a benchmark using $Na_2Ca_3Al_2F_{14}$ data collected in five detector banks of WISH at ISIS, refinement of 193 parameters took about 3 seconds with crysta compared to 20 seconds with FullProf, using the same initial values. All refined structural model parameters agreed with FullProf within 0.3 combined standard uncertainties. These timings were measured on the default six-core VISA Linux virtual machine used as a remote desktop for ESS users.

Current work includes support for magnetic structure refinement for DREAM. Polarised-neutron analysis for MAGIC is one of the next main priorities. A crysta-based web application is available for testing without installation, although it is still in early development. Feedback and more examples from instrument teams and users can help guide further development.

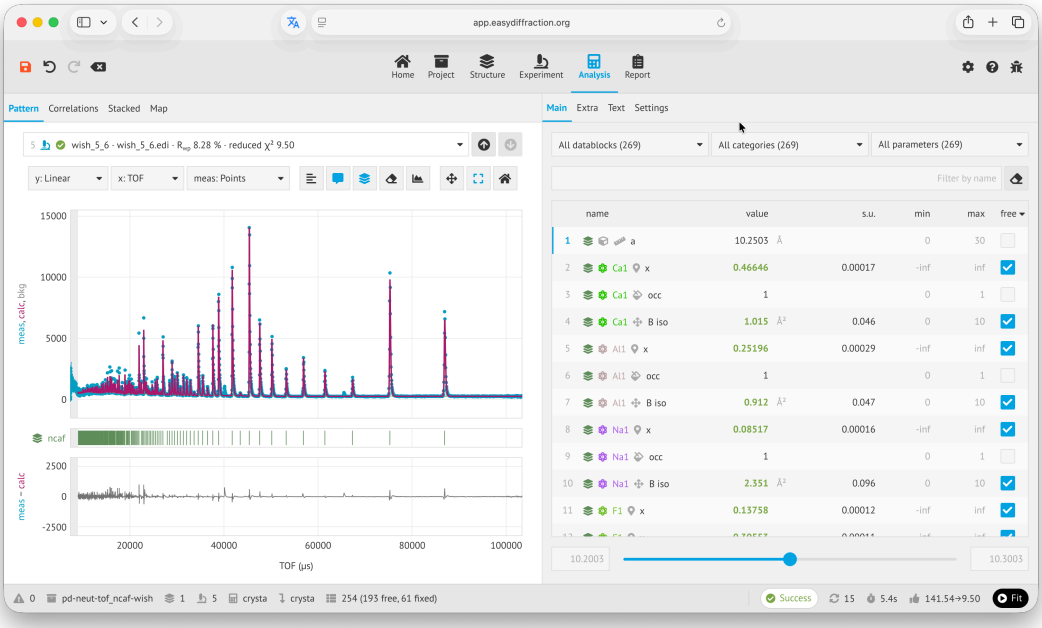


Figure 1: Try the EasyDiffraction web application

Topic

Crystallography, in-operando studies

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